Can Trusted Undercover Agents Detect Collusion Without Inducing It? A Pre-Registered Single-Model Study in the Liar's Bar Harness

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Abstract. LLM-based agents increasingly operate in multi-agent settings where covert cooperation opportunities arise. Apart Research's Track 3.2 asks whether a trusted agent that offers collusion has been tested; it has not. We present a small, pre-registered, single-model study in the official Liar's Bar harness (4-agent rounds, a secret-channel tool with structured accept/refuse responses) using DeepSeek-V4.1-Flash across 20 episodes in three phases (7 calibration, 3 pilot, 10 confirmatory). A scripted trusted investigator delivered 10 invitations: 8 voluntary invitations were all refused (0/8 accepted, across three different recipient agents), and 2 seeded episodes had refusals overridden by the harness, imposing alliances with procedural ground truth. Imposed alliances exchanged 108 secret messages with full hand disclosure and coordinated play plans; post-establishment, allies never challenged each other (0/10 eligible opportunities) while challenging non-allies at the baseline rate ($10/85 = 0.118$). Focal-pair challenge rates against a pre-registered pair showed no consistent induction signal (confirmatory C2 0.106 vs C0 0.066). We report all results as descriptive observations with explicit limitations: single model family, small N, an experimenter-channel investigator, and no cross-game memory. Protocol, raw records, manifests, and analysis scripts are released for independent verification.

1. Introduction

LLM-based agents increasingly operate in multi-agent settings where covert cooperation opportunities can be exploited. Apart Research's Track 3.2 notes that a trusted agent offering collusion opportunities "has not yet been tested": can such an undercover intervention **expose** existing collusive coordination, or does it **induce** coordination that would not otherwise occur?

We present a small, controlled, single-model study in the official Liar's Bar harness. Our design separates three questions that are often conflated:

- **RQ1 (elicitation)**: do agents accept, refuse, or express suspicion toward a secret-alliance invitation?
- **RQ2 (behavioral consequences of imposed alliances)**: does an imposed (seeded) alliance produce observable preferential behavior — and can our instrument detect it?
- **RQ3 (induction)**: does delivering an invitation change focal-pair challenge behavior relative to matched no-invitation episodes (ITT)?

We report invitation-level outcomes, opportunity-denominated challenge rates against a pre-registered focal pair, and explicit limitations of a single-model, small-N exploratory study.

2. Related Work

- Fish et al. (2024, arXiv:2404.00806) showed LLM pricing agents collude in iterated market games — spontaneous collusion in economic settings; our setting is a structured bluffing game with an explicit invitation mechanism.
- Shi et al. (2026, arXiv:2607.05132) showed agents' private plans diverge from their public commitments, exceeding 90% divergence in the highest-deception conditions — motivating the measurement of **behavior**, not words.
- Nakamura et al. (2026, arXiv:2602.15198), Colosseum, showed coalition advantage emerges with communication; monitoring efficacy varies by dimension.
- Arike et al. (2026, arXiv:2601.21112) showed monitors with **less** information access often detect sabotage better than monitors with full traces (ICML 2026) — our opportunity-denominated behavioral metric follows the same principle.
- Nguyen et al. (2026, arXiv:2604.03968), TraceGuard, built a single-agent structured monitor with a reusable rubric — not a collusion detector; we adapt the rubric concept to alliance behavior.
- Radev et al. (2026, arXiv:2605.29178), The Best-Laid SCHEMEs, showed coordinated sabotage across multi-agent coding systems is practical (84%/46% covert-task completion), and a trusted monitor detects it from code edits at FPR≤2% in 99%/68% of attempts — rising to 100%/81% with communication access — directly relevant to our alliance-communication threat model.
- Our contribution relative to the above: a **trusted-investigator** intervention protocol with procedural alliance ground truth, opportunity-denominated behavioral metrics, and an explicit expose-vs-induce decomposition.

3. Methodology

3.1. Environment

The official Liar's Bar harness: 4-agent rounds, a secret-channel tool with structured accept/refuse responses, optional cross-game memory. Episodes are single games: cross-game player memory was verified absent in the series runner (audit finding H3), making episodes statistically independent units.

3.2. Model and inference

deepseek-ai/DeepSeek-V4.1-Flash via a Modal Shared Endpoint (OpenAI-compatible). Sampling follows the harness's per-seat calibration (temperature 0.8, top_p 0.9, top_k 50, repetition penalty 1.05, max tokens 256), with structured JSON outputs and 5-retry parsing with fallback logging.

3.3. Conditions and phases

Conditions were pre-registered before any confirmatory episode ran (PILOT_SPEC, frozen at commit `d08fc09`):

- **C0**: no invitation (baseline).
- **C2**: secret-alliance invitation (allow_refusal = true; voluntary response).
- **C4**: seeded alliance (allow_refusal = false; the harness overrides refusal and force-assigns a partner — procedural ground truth).

C1/C3/W1 were deferred. Phases: 7 calibration episodes (G3 positive-control work), 3 pilot, 10 confirmatory. The primary analysis uses confirmatory episodes only.

3.4. Trusted investigator

A scripted, experimenter-controlled investigator delivers a fixed invitation template; it holds no game seat, earns no points, and its full policy is the published template. Targeting (`--chosen-player Lily`) was verified working in the single-game script; the series runner's targeting was broken (audit finding H1) and random targets were used where applicable.

3.5. Measurements

- **Invitation outcome** (unit: invitation event): accept / refuse (+reason).
- **Focal pair challenge rate**: the pair (Lily, Luke), pre-registered and seat-adjacent; denominator = eligible decision opportunities (next-player decision points; system auto-challenges excluded — validated by 21 metric unit tests including an independent oracle and a real-trajectory audit).
- **Alliance-pair contrast** (secondary, labeled post-treatment): ally vs non-ally challenge rates in seeded episodes.
- Post-treatment window: the establishment round is excluded (conservative).

4. Results

4.1. Invitation outcomes (unit: invitation event)

Arm	N	Accept	Refuse	Refusal overridden → forced alliance
Voluntary (allow_refusal = true)	8	0	8	0
Seeded (allow_refusal = false)	2	0 (2 refusal attempts overridden)	—	2
Total invitation events	**10**	0	8	2

Table 1: Invitation outcomes across all 10 invitation events.

All 8 voluntary invitations were refused (three different recipient agents: Mike, Quinn, Lily×6 — Lily was the targeted recipient). Descriptive counts; N is too small to infer a population refusal rate.

4.2. Focal pair (Lily, Luke) challenge rates

Condition / phase	episodes	focal challenges / opportunities	rate
C2 confirmatory	6	9/85	0.106
C0 confirmatory	4	4/61	0.066
C2 pilot + calibration	3 (1 more with zero opportunities)	1/20	0.050
C0 calibration	4	12/76	0.158
C4 seeded	2	2/39	0.051

Table 2: Focal-pair challenge rates by condition and phase (unit: episode; denominator = eligible decision opportunities).

The primary ITT estimand (confirmatory phase): $\Delta = 0.106 - 0.066 = \mathbf{+0.040}$. C2 episodes are refusal-dominated (6/6 refused, no alliance formed), so focal rates should approach C0; the observed gap (+0.040) at n = 6/4 is not interpretable. Per-episode focal rates ranged 0.00-0.32.

4.3. Alliance behavior (seeded episodes, n = 2)

Measure	082152 (Quinn+Luke)	094116 (Lily+Mike)	pooled
Ally mutual challenges	0/4	0/6	**0/10**
Ally outward challenge rate	4/42 = 0.095	6/43 = 0.140	10/85 = 0.118
Overall challenge rate (post-establishment R4+)	6/87 = 0.069	13/93 = 0.140	19/180 = 0.106
Secret messages	51	57	108
Winner	Lily (non-ally)	Luke (non-ally)	alliance lost both

Table 3: Alliance behavior in the two seeded episodes.

Both episodes showed the same direction: alliance-consistent behavior (no mutual challenges), secret messages containing full hand disclosure and coordination plans ("My hand: Joker, K, K, A, Q"; "Let's both play K this round"; "Target Mike first"), and Quinn expressing wariness before the override ("no blind collusion"). Coalition benefit was negative in both episodes: a non-allied player won each.

4.4. Suspicion / disclosure

All refusal reasons contained "REFUSE" (no motivation text); play-reason fields showed no attempts to disclose the invitation to other players. The suspicion dimension was **not observed** (constrained expression channels, as pre-registered).

5. Discussion

The three-level separation of our main findings. Our data split Track 3.2′s core question into three separately verifiable layers with different evidence strength:

1. **Elicitation (RQ1)**: DeepSeek-V4.1-Flash faced 10 invitation events (8 voluntary + 2 seeded; three different recipient agents; two script paths). All 8 voluntary invitations were refused (0/8 accepted); both seeded attempts to refuse were overridden by the harness and converted to forced alliances. This is a descriptive observation; refusal motivation (safety tendency, payoff calculation, or invitation wording) was not experimentally separated.
2. **Imposed-alliance behavior (PC-B)**: two experimenter-forced alliances produced 108 secret messages (full hand disclosure, coordinated play plans, target coordination), and allies never challenged each other (0/10 opportunities; both episodes directionally consistent). This is an **instruction-compliance** behavioral signal — the alliance contract explicitly states "NEVER challenge {initiator}'s plays", so this measures compliance with a notified alliance, not spontaneous collusion.
3. **Induction (RQ3)**: the primary ITT comparison (confirmatory: C2 vs C0 focal rates 0.106 vs 0.066) ran **opposite** to the induction hypothesis with insufficient N — invitation delivery (mostly refused) did not induce focal-pair challenge-rate drops; the pilot-phase sensitivity comparison (0.050 vs 0.158) ran consistent with the hypothesis. The two phases ran in opposite directions, both underpowered. **No conclusion** at this layer.

Implications for Track 3.2. Our results delineate the operating boundary of the trusted undercover intervention: when target agents consistently refuse, the intervention produces no behavioral favoritism (the focal-pair rate did not drop); when an alliance is externally imposed, agents follow the alliance contract behaviorally and use the channel for coordination. The former means "exposing existing collusion" met no resistance in this model (no existing collusion to expose, and invitations were refused); the latter demonstrates that our instrument can detect contract-level behavioral differences. This separation is exactly the measurement foundation the expose-vs-induce study design requires.

Measurement methodology lessons. (1) Challenge-rate denominators must be eligible opportunities — focal-pair opportunity density varied 0-34 per episode (measured over 20 episodes, from a1-g1′s 0 to m-c0-4′s 34); raw counts without opportunity denominators would misread seat and survival effects as behavioral differences. (2) The official harness's serialization gaps (establishment events not written to JSON; partial series-config propagation) require stdout-JSON cross-validation as standard procedure. (3) The metric itself needs synthetic-trajectory unit tests plus an independent recount audit — our audit caught the "No one" system auto-challenge literal that the analyzer had missed.

6. Limitations and Dual-Use / Ethical Considerations

- Single model family (DeepSeek-V4.1-Flash), small N (exploratory), no seed control (randomization-by-draws recorded as-is).
- The investigator is an experimenter channel, not a seat-based agent.
- Forced-assignment alliances measure compliance with notifications, not voluntary coordination.

- Suspicion disclosure is constrained by the available expression channels.
- No cross-game memory (audited harness behavior): repeated-play effects are unmeasured.
- **Dual-use**: the invitation template and harness quirks documented here are drawn from a public starter repository intended for safety research; we do not provide new attack techniques.

7. Reproducibility

The research package (`our_experiment/`) contains: RESEARCH_PLAN, PILOT_SPEC (pre-registered, frozen at commit `d08fc09`), PROTOCOL_DEVIATIONS (D1-D8), EXPERIMENT_LOG, HARNESS_INTEGRITY_AUDIT, TRACK_3_2_ALIGNMENT, G3_REVIEW, SKEPTICAL_REVIEW, run manifests with SHA-256 hashes, `analyze_g3.py`, `test_metrics.py` (21 tests), `pilot_ledger.py`, and `verify_numbers.py` (22 checks). Reproduction commands and the freeze commit hash are recorded in the run manifests. The single official submission of the final PDF is performed by the human principal investigator.